Prob. 5. Use Leibniz rule:

$$(1) \quad \frac{d}{dx}\int_{\alpha(x)}^{\beta(x)} f(x,t)\,dt = \int_{\alpha(x)}^{\beta(x)} \frac{\partial f(x,t)}{\partial x}\,dt + f(x,\beta(x))\cdot\beta'(x) - f(x,\alpha(x))\cdot\alpha'(x)$$

to compute the followings: (Simplify as much as possible) Also, be sure to simplify to a single fraction. If you want to know if you obtained the correct answer, check by evaluating the derivative of the integral (i.e. not using Leibniz rule):

(a)

$$(2) \rightarrow \frac{d}{dx}\int_0^{x^3} \frac{dt}{x^2+t} =$$

With Leibniz rule: $(x^2+t)^{-1}$

$$\frac{d}{dx}\int_0^{x^3} \frac{dt}{x^2+t} = \int_0^{x^3} \frac{\partial}{\partial x}\left[\frac{1}{x^2+t}\right]dt + \frac{1}{x^2+x^3}\cdot(x^3)' - \frac{1}{x^2+0}\cdot(0)' =$$

$$= \int_0^{x^3} -\frac{2x}{(x^2+t)^2}\,dt + \frac{3x^2}{x^3+x^2} - 0 = +\int_0^{x^3} \frac{2x}{(x^2+t)^{-2}}\,dt + \frac{3x^2}{x^2(x+1)} =$$

$$= +\left[2x\cdot\frac{1}{x^2+t}\right]_{t=0}^{t=x^3} + \frac{3}{x+1} =$$

$$+\left[2x\cdot\left(\frac{1}{x^2+x^3} - \frac{1}{x^2+0}\right)\right] + \frac{3}{x+1} =$$

$$= 2x\left[\frac{1}{x^2(x+1)} - \frac{1}{x^2}\right] + \frac{3}{x+1} = +2x\cdot\left[\frac{-1}{x(x+1)}\right] + \frac{3}{x+1} =$$

$$(3) \quad = \frac{-2}{x+1} + \frac{3}{x+1} = \frac{3-2}{x+1} = \frac{1}{x+1}.$$

Without Leibniz rule:

$$\frac{d}{dx}\int_0^{x^3} \frac{1}{t+x^2}\,dt = \frac{d}{dx}\Big[\ln(x^2+t)\Big]_{t=0}^{t=x^3} =$$

$$= \frac{d}{dx}\Big[\ln(x^2+x^3) - \ln(x^2+0)\Big] = \frac{d}{dx}\Big[\ln(x^2+x^3)\Big] - \frac{d}{dx}\Big[\ln(x^2)\Big]$$

$$= \frac{1}{x^2+x^3}\cdot(2x+3x^2) - \frac{1}{x^2}\cdot(2x) =$$

$$= \frac{2+3x^{\backslash x}}{x+x^2} - \frac{2^{\backslash x+x^2}}{x} = \frac{2\cancel{x}+3x^2-2\cancel{x}-2x^2}{x(x+x^2)} = \frac{x^2}{x\cdot x(1+x)} =$$

(4) $$= \frac{1}{1+x}.$$ This matches the result in (3), as desired.

(6) Using Leibniz rule:

(5) $$\frac{\partial}{\partial x}\int_0^x \sqrt{1+y^3t^2}\,dt = \int_0^x \overbrace{\frac{\partial}{\partial x}\Big[\sqrt{1+y^3t^2}\Big]}^{=0 \text{ since inside is not a fun'n of } x.}\,dt +$$

$$+ \sqrt{1+y^3\cdot(x)^2}\cdot\overbrace{\frac{d}{dx}(x)}^{=1} - \sqrt{1+y^3\cdot(0)^2}\cdot\overbrace{\frac{d}{dx}(0)}^{=0}$$

$$= \int_0^x 0\,dt + \sqrt{1+y^3\cdot x^2}\cdot 1 - 0 = \sqrt{1+y^3x^2}.$$

(I thought it might be difficult to check without Leibniz, so compared the result with a classmate.)